

COVID-19: STEMI and VF Arrest

Section 1: Case Summary

Scenario Title:	COVID-19: STEMI and VF Arrest
Keywords:	COVID, STEMI, Ventricular Fibrillation, Cardiac Arrest
Brief Description of Case:	Ronda is a 50 yo patient who presents with 1.5 hours of typical cardiac chest pain. She has known COVID positive contacts and mild URTI symptoms. The ECG shows an anterior STEMI. The team will need to perform initial work-up and management of a patient with STEMI. While this is occurring, the patient suffers a VF arrest. The team will need to go through the ACLS algorithm while taking all precautions required in caring for a patient with suspected COVID.

Goals and Objectives	
Educational Goal:	To practice effectively caring for a COVID positive patient presenting with STEMI and VF arrest while protecting the safety of the healthcare team.
Objectives: (Medical and CRM)	1. Recognize a COVID risk patient and sequester and protect team with appropriate PPE 2. Recognize and treat a STEMI that unexpectedly progresses to VF arrest 3. Demonstrate effective communication and resource allocation in a COVID suspected patient requiring urgent intervention 4. Demonstrate effective donning and doffing of PPE 5. Utilize protective measures in a suspected COVID patient in cardiac arrest
EPAs Assessed:	N/A

Learners, Setting and Personnel			
Target Learners:	☐ Junior Learners	☐ Senior Learners	☒ Staff
	☒ Physicians	☒ Nurses	☒ RTs
	☐ Other Learners:		
Location:	☒ Sim Lab	☒ In Situ	☐ Other:
Recommended Number of Facilitators:	Instructors: 1		
	Confederates: 0		
	Sim Techs: 1		

Scenario Development	
Date of Development:	March 29, 2020
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Ronda Smith		Age: 50		Gender: F	Weight: 70 kg
Presenting complaint: Chest pain					
Temp: 36.8 °C	HR: 88/min	BP: 148/80	RR: 18/min	O ₂ Sat: 98%	FiO ₂ : RA
Cap glucose: 7 mmol/L			GCS: 15 (E4 V5 M6)		
Triage note: 50 yo female with 1.5 hours of chest pain and L arm heaviness. 2 days of mild URTI symptoms. Contact with COVID patients at LTC home, where the patient works. EKG not done at triage.					
Allergies: NKDA					
Past Medical History: HTN Hypercholesterolemia			Current Medications: Lipitor 40 mg daily Ramipril 5 mg daily		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, confederate, SP, etc.)?</i>	
Patient (mannequin's voice) gives following information if asked by learners: Ronda is a 50 yo patient who presents with 1.5 hours of typical cardiac chest pain. She has known COVID positive contacts and mild URTI symptoms. She works as a nurse at a LTC home. She has had a cough for 2 days. She also has a runny nose. She does not have hemoptysis. It is hard to take a deep breath. She had subjective chills yesterday. She has been self-isolating for the last 24 hours.	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: normal heart sounds and pulses	Neuro: normal
Resp: good a/e bilaterally, no crackles or wheeze	Head & Neck: normal
Abdo: soft, non-tender	MSK/skin: diaphoretic
Other:nil	

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>) – adult, computerized mannequin
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
PPE: gown, N95 mask, faceshield, gloves Nasopharyngeal swab Defibrillator and pads LMA Intubation materials, including viral filter HEPA filter for room
C. Required Medications
ASA, nitro spray, ticagrelor/clopidogrel, opioids/pain medication, heparin ACLS medications
D. Moulage
Diaphoresis spray
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> A – Airway protected, patient talking. B – Spontaneous breathing, hard to take a deep breath. Good air entry bilaterally, no crackles or wheezes on auscultation. No cyanosis. C – Extremities well perfused. Pulses equal bilaterally. D – GCS 15, says feeling lightheaded with generalized weakness E – Skin is diaphoretic throughout. No signs of trauma or injury on head-to-toe examination.

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Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Charge RN	"I just put a woman with chest pain into the negative pressure resuscitation room." "She has had 1.5 hours of severe chest pain and L arm heaviness. She looks pretty unwell and is diaphoretic. I haven't done an ECG yet." "She's an RN from a long-term care home and apparently has been in contact with some COVID patients. She says she does have some mild URTI symptoms as well for the past couple of days"
Patient	When learner engages the patient, she says: "The pain is so bad, its 10/10, right in the middle of my chest" "It is getting worse, and I'm feeling lightheaded"

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus HR: 88 BP: 108/70 RR: 18 O ₂ SAT: 98% T: 36.8°C GCS: 15	Complains of 10/10 chest pain and progressive lightheaded	<u>Expected Learner Actions</u> ☐ Don droplet PPE ☐ Minimize staff in room ☐ Obtain bloodwork/viral swab ☐ Cardioresp monitors ☐ Obtain EKG ☐ STEMI medications ☐ Call cath lab/cardiology	<u>Modifiers</u> - EKG provided when asked for <u>Triggers</u> - STEMI meds are given, or after 6 minutes → Initial Mgmt	As time goes on with the patient in this state, she will become more drowsy, until she becomes unresponsive in state 2.
2. VF Arrest – Initial Mgmt Rhythm: VF HR: -- BP: -- RR: --	Pt unresponsive and pulseless	<u>Expected Learner Actions</u> ☐ Call for help ☐ Place facemask on patient ☐ Place pads on patient ☐ Defibrillate VF rhythm ☐ Leave room	<u>Modifiers</u> - Rhythm check → VF <u>Triggers</u> - Team Re-enters → Ongoing Mgmt	
3. VF Arrest – Ongoing Mgmt Rhythm: VF HR: -- BP: -- RR: --	Pt unresponsive and pulseless	<u>Expected Learner Actions</u> ☐ Team dons airborne PPE ☐ Start chest compressions ☐ Epinephrine ☐ NRB or BVM on airway ☐ Avoid bagging ☐ Hold CPR for airway mgmt ☐ Place ETT or LMA	<u>Modifiers</u> - Rhythm check → VF - Resistant to 2 shocks <u>Triggers</u> - Defibrillate x 3 → ROSC	
4. ROSC Rhythm: Sinus HR: 92 BP: 105/60 RR: 10 O ₂ SAT: 92%	Pulses present GCS 2T	<u>Expected Learner Actions</u> ☐ Complete STEMI meds ☐ Repeat ECG ☐ Prepare for transfer with appropriate COVID precautions ☐ Post-rosc care (Foley, sedation, cooling) ☐ PPE doffing with safety officer	<u>Modifiers</u> - Cath lab calls, ready for patient <u>Triggers</u> PPE is doffed → END CASE	



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Appendix A: Laboratory Results

Labs will not be back before the case ends.

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!

ECG showing anterior STEMI



<https://litfl.com/anterior-myocardial-infarction-ecg-library/>

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

NB: This is controversial and there are many potential ways to approach code blue in the suspected or confirmed COVID-19 patient. This case is based upon the [covidcriticalcare.ca](https://www.covidcriticalcare.ca) algorithm.

Here are two other algorithms for reference and discussion

- http://www.bccdc.ca/Health-Info-Site/Documents/COVID_Adult_CPR_Protocol.pdf
- <https://www.resus.org.uk/media/statements/resuscitation-council-uk-statements-on-covid-19-coronavirus-cpr-and-resuscitation/covid-healthcare/>

Objectives:

- Recognize a COVID risk patient and sequester and protect team with appropriate PPE
- Recognize and treat a STEMI that unexpectedly progresses to VF arrest
- Demonstrate effective communication and resource allocation in a COVID suspected patient requiring urgent intervention
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Key moments to reflect on:

- Identification of possible COVID positive patient and considerations to room placement
- Utilization of appropriate PPE
- Initial management of STEMI
- Provider response to cardiac arrest
- Changes to routine ACLS care during a “Protected Code Blue”
- Preparing COVID patient for transfer

Sample questions for debriefing:

- Who should be present in the resuscitation room for these patients? How do we balance having the necessary personnel present while mitigating risk to the healthcare team/exposure?
- What personal protective equipment (PPE) must providers wear?
- What are next steps for staff members who may have been inadvertently exposed?
- What are the steps in decontaminating the room and equipment after a patient encounter such as this?
- What are the considerations for transfer of these patients to another department, such as the Cath Lab?

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PROTECTED CODE BLUE

(FOR PRE-ARREST AND ARREST PATIENTS)

KEY PRINCIPLES

1 Rapid Team huddle before entering room

Roles, responsibilities, equipment, medications, PPE.



2 Minimize number of people in room



**3 Don Appropriate PPE before entering
NO EXCEPTIONS**



FOR AIRWAY (PRE-ARREST)

→ 1 MD / 1 RN / 1 RT

FOR CARDIAC ARREST (CPR)

→ 1 MD / 2 RN / 1 RT

IF AVAILABLE, NEGATIVE PRESSURE ROOM OR ROOM WITH HEPA FILTER AND CLOSED DOORS

AIRWAY & INTUBATION (see intubation checklist on Code Cart)

- 1 Bagging – ONLY for severe desaturation in non-cardiac arrest with ALL team members in appropriate PPE. (If bagging use 2 person technique for optimal seal and smallest possible tidal volumes.)
- 2 Assess airway – if expected difficult, page Anesthesia STAT. Wait for anesthesia if patient safety is not compromised.
- 3 Intubation: optimize first pass success: Rapid sequence (Ketamine 2mg/kg IV + Rocuronium 100-150 mg IV)
Video-laryngoscopy by most experienced operator
- 4 If first attempt failure: page Anesthesia STAT and consider LMA.

CPR

- 1 Prior to initiating CPR, don PPE and place procedure mask over patient's face.
- 2 2 RNs or delegate perform CPR in 2 minute rotations.
- 3 **NO** Bag-valve mask ventilation prior to intubation during CPR.
- 4 Control Airway: As soon as equipment is ready, pause CPR and intubate patient. Resume CPR once patient is intubated and BVM with filter is attached.

Team Members & Personal Protective Equipment



DONNING ORDER

- 1 HAND HYGIENE
- 2 LEVEL 4 GOWN (OR EQUIVALENT)
- 3 FIT-TESTED N95 MASK
- 4 EYE PROTECTION
- 5 GLOVES



SAFETY OFFICER
Coaching all donning and doffing

DOFFING ORDER

- 1 GLOVES
- 2 LEVEL 4 GOWN (OR EQUIVALENT)
- 3 HAND HYGIENE
- 4 EYE PROTECTION
- 5 FIT-TESTED N95 MASK
- 6 HAND HYGIENE



DEBRIEF
POST RESUSCITATION

22 March 2020

References

1. <https://covidcriticalcare.ca/cardiac-arrest/>
2. <https://www.sim-one.ca/community/news/covid-19-simulations>



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